

LOGIC FRIDAY For Combinatorial Digital Logic Design

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Digital electronics rarely leaves you, no matter where you go. Right from the small register in your controller to the multiplexer that routes the signals in your circuit, it is present in almost everything you handle. The subject being a delight for some and a nightmare for the others, this article introduces a tool that helps make your digital logic design experience easy and pleasurable. Have a look at Logic Friday.

Visualise your logic function in totality

The logic of any combinatorial system design is always analysed for all possible input combinations, the different states its output can take, an expression for every possible output and, the most favourite of all, the logic diagram. Logic Friday lets you ease over the process by simultaneously viewing and accessing four windows: functions, truth table, equations and gate diagrams. When you make a modification in one window, the change is reflected in the others automatically. Every function includes a detailed description, right from the number of inputs and outputs to the number of min terms that make it true.

Espresso for efficient minimisation

Logic Friday takes the help of

Espresso logic minimiser to efficiently reduce the functions in your electronic design. Instead of using the traditional Karnaugh map method of min term reduction, the program

manipulates the function iteratively to give a closely approximated result, eliminating redundancy. The same result is optimised by factorisation and mapped onto basic cells

in the technology library, while dealing with field programmable gate arrays and application specific integrated circuits. This method of minimisation turns out to be efficient, using less memory and computation time.

You have options of performing a fast or an exact minimisation. The former would give you a good result that may not be truly minimal, while the latter takes a significant amount of time to give you the perfect result.

Also, for functions with multiple outputs, you can choose to jointly optimise the total number of product terms or independently optimise each min term. The second is useful while mapping to gates, as you will end up with a simpler gate diagram, but might result in a large number of terms in the resulting function.

The function explained as gates

There is no digital electronics without a gate diagram. Universal gates NAND and NOR help represent any complex logic, and one of the two will have to be specified as the type for

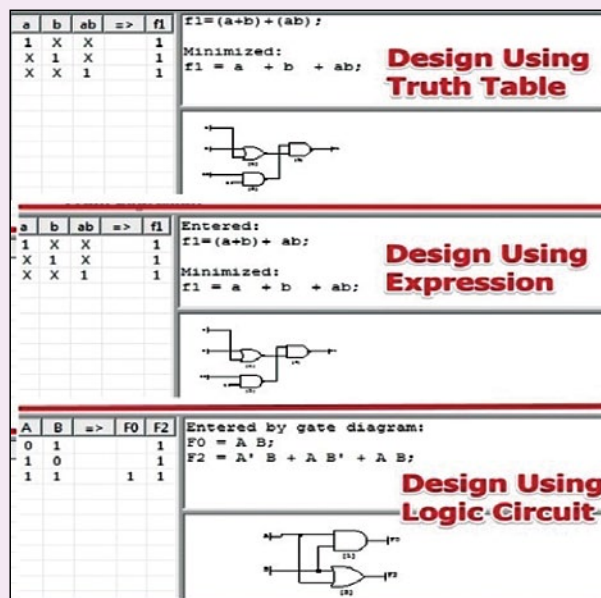


Fig. 1: Designing with Logic Friday
(Image courtesy: www.ilovefreesoftware.com)

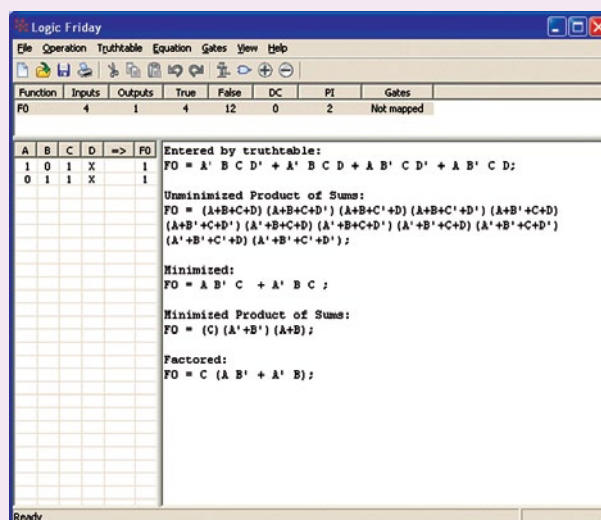


Fig. 2: Manipulating equations